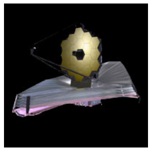

3038 - An archival study of AGN host galaxies with JWST NIRCam imaging

Cycle: 2, Proposal Category: AR

INVESTIGATORS

Name	*Institution*
Dr. Mingyang Zhuang (PI)	**University of Illinois Urbana-Champaign**
Dr. Yue Shen (CoI) (CoPI) (US Admin CoI)	University of Illinois at Urbana - Champaign
Dr. Junyao Li (CoI) (CoPI)	University of Illinois at Urbana-Champaign

OBSERVATIONS

ABSTRACT

JWST's unparalleled imaging capabilities are starting to revolutionize the field of AGN host galaxy studies at high redshift. As JWST rapidly accumulates NIRCam imaging data across extragalactic fields, it is essential to understand the systematics of the instrument and develop efficient and robust methodologies for high-z AGN host measurements. This program will utilize archival NIRCam imaging data to perform a comprehensive analysis of the point-spread-function (PSF) as functions of field location/rotation, filters, dither patterns and star properties, and apply the results to AGN host measurements in well-covered extragalactic fields. The main deliverables from this program include: (1) detailed PSF characterization of NIRCam imaging and optimal strategies to construct robust PSF models for AGN host studies; (2) detailed simulations of mock observations to quantify the caveats and limitations of host measurements in high-z (broad-line) AGNs with NIRCam imaging; (3) measurements of host galaxy stellar properties for type 1 AGNs across much of the cosmic history to investigate the low-to-intermediate mass range of the black hole mass - host stellar mass relation, and to systematically discover and characterize the populations of galactic-scale dual and offset AGNs; (4) a detailed comparison of type 1 and type 2 AGN hosts up to z~3. The technical development and science results from this program will build solid ground to capitalize on the unique capabilities of NIRCam imaging, and enable broader science in the field of AGN and galaxy evolution.

OBSERVING DESCRIPTION

This AR proposal will use all archival NIRCam imaging data for PSF characterization, and focus on the COSMOS field and S-CVZ field to study AGN host galaxies with NIRCam data. As of Jan 20, 2023, there are already a large number of NIRCam imaging programs public avaliable in the archive, and the program will continue building the statistics with upcoming data (e.g., from the COSMOS-Web treasury program). These NIRCam imaging observations cover all ranges of observing parameters and field properties, and will enable a comprehensive study of the NIRCam PSF, as well as science applications to a large sample of AGNs across cosmic time.